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Water Resources Research

RESEARCH ARTICLE
10.1029/2022WR033342

# Is It Possible to Quantify Irrigation Water-Use by Assimilating a High-Resolution Satellite Soil Moisture Product?

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**Key Points:**

- Complex human irrigation decisions can be modeled using the particle batch smoother (PBS) approach
- Irrigation can be estimated accurately by assimilating a weekly high-resolution soil moisture data containing the irrigation signal
- Instead of using the irrigation module in the land surface model, PBS can be used to account for irrigation

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**Citation:**

Jalilvand, E., Abolafia-Rosenzweig, R., Tajrishy, M., Kumar, S. V., Mohammadi, M. R., & Das, N. N. (2023). Is it possible to quantify irrigation water-use by assimilating a high-resolution satellite soil moisture product? *Water Resources Research*, *59*, e2022WR033342. https://doi.org/10.1029/2022WR033342

Received 28 JUL 2022
Accepted 20 MAR 2023

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**Abstract** Irrigation is the largest human intervention in the water cycle that can modulate climate extremes, yet irrigation water use (IWU) remains largely unknown in most regions. Microwave remote sensing offers a practical way to quantify IWU by monitoring changes in soil moisture caused by irrigation. However, high-resolution satellite soil moisture data is typically infrequent (e.g., 6–12 days) and thus may miss irrigation events. This study evaluates the ability to quantify IWU by assimilating high-resolution (1 km) SMAP-Sentinel 1 remotely sensed soil moisture with a physically based land surface model (LSM) using a particle batch smoother (PBS). A suite of synthetic experiments is devised to evaluate different error sources. Results from the synthetic experiments show that unbiased simulations with known irrigation timing can produce an accurate irrigation estimate with a mean annual bias of 0.45% and a mean $R^2$ of 0.97, relative to observed IWU. Unknown irrigation timing can significantly deteriorate the model performance, resulting in an increased mean annual bias to 23% and decreased mean $R^2$ to 0.36. Adding random noise to synthetic observations does not significantly decrease model performance except for the experiments with low observation frequency (>12 days). In real-world experiments, the PBS data assimilation approach underestimates observed IWU by 18.6% when the timing of IWU is known. IWU estimates are consistently significantly higher over irrigated pixels compared to the non-irrigated pixels, indicating data assimilation skillfully conveys irrigation signals to the LSM.

**Plain Language Summary** Determining and modeling the amount of water applied as irrigation for agriculture is a very difficult task. Traditionally, it is measured in situ, and in most parts of the world such observations are not available or the local agencies do not share the data due to many socio-economic/political reasons. Studies have shown that signal of irrigation over an agricultural field can be observed through a high-resolution microwave remote sensing technique. However, the determination of the quantity of irrigation applied to the agricultural field without in situ measurement is very challenging. This study demonstrates a possible way to quantify irrigation through modeling and assimilating microwave remote sensing-based soil moisture data. The results of this study show significant skills to quantify the amount of water used for irrigation. Such study is essential for water resource management and budgeting.

## 1. Introduction

Agricultural production is projected to require a 70% expansion by 2050 as a result of population growth, climate change, and dietary shifts toward water-intensive products associated with increasing incomes (Tilman & Clark., 2015). Food production is mainly sustained by irrigation (Jägermeyr et al., 2015), which is by far the largest consumer of freshwater resources globally (Döll & Siebert, 2002). However, the planetary limit for freshwater withdrawal is quickly approaching (or already exceeded) in many parts of the world (Steffen et al., 2015). Moreover, it is expected that 25%–40% of water for irrigation agriculture will be reallocated to higher productivity sectors in the near future (World Bank, 2022). Thus, authorities seek to limit agricultural water consumption globally. However, a key requirement to enforce any regulation is monitoring the water withdrawn by farmers (Foster et al., 2020). Monitoring groundwater storage changes through GRACE (Gravity Recovery and Climate Experiment) satellite observations has revealed significant depletion in major aquifers across the globe (Frappart & Ramillien., 2018; Joodaki et al., 2014; Scanlon et al., 2012), mainly due to unsustainable water withdrawal

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(consuming more water than received from rainfall) for agriculture (Döll et al., 2014). Thus, quantifying Irrigation water use (IWU) is a valuable component for the optimal management of depleting surface and groundwater resources to sustain crop production (Abbot et al., 2019).

Despite the importance of monitoring agricultural water use, much of the water withdrawn for irrigation is unmetered due to the high installation and maintenance costs of ground sensors (Wells, 2015). For instance, in the United States (U.S.) less than 36% of wells are equipped with a flowmeter (Foster et al., 2020) and the fraction is much less in developing countries where smallholder agriculture is the common practice (Shah, 2014; Wester et al., 2009). Even when an in situ metering network is in place, there is usually a large gap between the formal policies and the local practices. Farmers are reported to adopt tactics to circumvent existing limitations such as illegal surface water diversion, drilling unlicensed wells, or tampering with meters installed on legal wells (al Naber & Molle, 2017; Balasubramanya & Stifel, 2020).

Satellite observations provide a unique opportunity to monitor irrigation and address many of the above-mentioned issues with in situ measurements. Thermal and optical sensors can measure evapotranspiration (ET) using crop coefficients obtained from satellite-based vegetation indices (Allen et al., 2005; Farg et al., 2012) or surface energy balance that is highly correlated with irrigation (Allen et al., 2007; Bastiaanssen et al., 2014; Filippelli et al., 2022; Javadian et al., 2019; Karimi et al., 2013). However, ET only measures consumptive water use which is generally less than the amount of irrigation applied to the field.

Irrigation water requirement can be estimated through a soil water deficit model that assumes irrigation is only triggered when soil moisture is below a certain threshold (the refill point) that is a function of the crop rooting depth and soil hydrologic properties and will be continued until the soil moisture reaches the field capacity (FC) such that the plant does not experience water stress or over-irrigation (Chai et al., 2016). Bretreger et al. (2022) used a constellation of Landsat data to estimate crop growth stage and crop coefficient and later used this coefficient in a water deficit model to quantify the IWU. A conceptually similar approach was adopted by Sadri et al. (2022) to predict the soil water deficit (needed irrigation) with up to a 14-day lead time using several near-real-time satellite observations and precipitation prediction in a hybrid physical-statistical machine learning model (FarmCan). However, in practice, a farmer's decision to start or end the irrigation does not necessarily follow the soil moisture status. For instance, a recent United States Department of Agriculture (USDA) survey in 2017 showed that 75% of farmers irrigated their fields based on a rule of thumb procedure, such as crop calendar, visual observations, and "what are neighbors doing!" (Zhang et al., 2021). Another study showed that farmers that have experienced loss due to water shortages in the past are more likely to irrigate more frequently (Balasubramanya et al., 2022). Thus, it is quite common for an agricultural plot to be over-or under irrigated.

Soil moisture (SM) measurements are needed to account for over- and under-irrigation. Microwave (MW) remote sensing can measure soil moisture effectively under all weather conditions and is proven to carry irrigation signals (Jalilvand et al., 2021; Lawston, Santanello, & Kumar, 2017). Brocca et al. (2018) and Jalilvand et al. (2019) have used satellite SM data with an SM-based inversion model (SM2RAIN model) to quantify irrigation water use. They relaxed some of the assumptions in the soil water deficit model to account for over-irrigation by allowing deep percolation to the lower soil layer and under-irrigation by adding a more sophisticated evapotranspiration module that allows water-limited evapotranspiration. They quantify the irrigation water use based on the changes in the MW satellite soil moisture data from Advanced Scatterometer (ASCAT) and Advanced Microwave Scanning Radiometer 2 (AMSR2). The results were consistent with in situ observations over a large and intensely irrigated area, but failed over small agricultural practices, mainly due to spatial mismatch between coarse-scale MW products (>25 km) and smallholder agricultural plots (<1 km). Despite the promising results from SM2RAIN obtained from these studies, the SM data itself remains a major challenge to this model, in that a higher resolution SM product is needed to detect the irrigation signal (Dari et al., 2020; Foster et al., 2020; Jalilvand et al., 2019; Massari et al., 2021; Modanesi et al., 2022; Zaussinger et al., 2019), while, increasing the spatial resolution corresponds with narrower sensor swath width and less frequent retrievals (Das et al., 2019). Dari et al. (2020, 2022) resolved the low temporal resolution issue by downscaling the coarse spatial but high temporal resolution data (e.g., Soil Moisture Active Passive (SMAP) Enhanced 9 km product) using methods such as DiSPATCh. However, as the downscaled soil moisture product uses the thermal observations from the Moderate Imaging Spectroradiometer (MODIS) platform that is affected by clouds and senses only the surface skin temperature, it might not capture the correct local SM dynamics within the topsoil, which is one of the most important factors in the success of SM-based inverted water balance models such as the SM2RAIN.

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Synergistic use of land surface modeling (LSM) and satellite data can address the issue of low frequency high-resolution SM observations. LSMs physically model land surface fluxes and state variables, including SM, by simultaneously solving surface water and energy budgets. LSMs may include an irrigation module to account for irrigation in the land surface processes (Leng et al., 2014, 2017; Ozdogan et al., 2010; Y. Pokhrel et al., 2012). However, the simplifying assumptions in the physics of the model, the errors in the input data, and the uncertainty associated with the model parameters and initial condition create large errors in LSM-simulated IWU (Modanesi et al., 2021; Puy et al., 2022). In situ and satellite data assimilation (DA) can help reduce part of the random errors in the model simulation and correct the model state based on the observations. For example, Felfelani et al. (2018) assimilated SMAP SM observations (using a 1-D Kalman filter) to constrain the target SM in the Community Land Model (CLM) irrigation module and significantly reduced the irrigation water requirement error over an intensely irrigated area in the U.S. Modanesi et al. (2022) used an Ensemble Kalman Filter (EnKF) to directly assimilate Sentinel 1 backscatter in co- and cross-polarization into the Noah-MP LSM with an irrigation scheme. They showed that assimilating Sentinel 1 backscatter in the VH polarization (that contains LAI information) can slightly improve irrigation simulations. Still, poor parametrization of the Noah-MP irrigation module does not allow the DA to improve the irrigation simulation significantly. Indeed, the irrigation module in most LSMs uses a demand-driven formulation that works similarly to the water deficit model and triggers irrigation when the root zone SM (RZSM) drops below a certain threshold and continues irrigating until reaching a target RZSM (Lawston, Santanello, Franz, & Rodell, 2017; Felfelani et al., 2018; Haddeland et al., 2006; Y. Pokhrel et al., 2012; Y. N. Pokhrel et al., 2016; Sorooshian et al., 2011). However, these models need information with a high level of uncertainty such as soil texture, soil hydrologic properties, crop type, fractional irrigated area (Puy et al., 2021), start and the end of the irrigation season (Massari et al., 2021), and irrigation efficiency (Puy et al., 2022). Moreover, they do not explicitly account for the human choices in irrigation scheduling and amount.

Abolafia-Rosenzweig et al. (2019) proposed a unique method, leveraging the particle batch smoother data assimilation algorithm, to quantify irrigation water use without reliance on an explicit irrigation module. In this method, the LSM is forced with precipitation superimposed with a range of irrigation, then satellite SM data which contains irrigation signals is assimilated with an ensemble of model simulations (particles) in a defined length of time (window), and particles are weighted based on the proximity to all satellite SM retrievals in that window. These weights can then be traced to the forcing to obtain the best estimate of water input (irrigation + precipitation) and by subtracting the observed precipitation, the amount of irrigation can be calculated. This method allows estimation of IWU with less reliance on presumptions about farmers' irrigation decisions, relative to standard irrigation modules, therefore, what truly happened on the ground and was captured by the satellite is used for estimation of the IWU. Furthermore, unlike EnKF or Ensemble Kalman Smoother (EnKS) techniques, the PBS does not assume the prior distribution of model states, making it a better fit for hydrological applications with non-Gaussian prior distributions (Lu et al., 2016; Yan & Moradkhani, 2016). All model states are also corrected in a physically consistent way and the weights from the analysis update can be used to directly update forcing (Abolafia-Rosenzweig et al., 2019; Margulis et al., 2015). One other advantage of this approach, relative to filtering DA techniques, is the batch assimilation of the SM observations, which makes it robust to the less frequent observations. Abolafia-Rosenzweig et al. (2019) showed that the assimilation of satellite retrievals with a 9-day return interval can estimate IWU with an $R^2$ larger than 0.85 relatives to a known amount of irrigation in synthetic experiments.

In this study, we assimilate SMAP-Sentinel 1 (SMAP-S1) 1 km SM data—that is proven to have the irrigation signal in both the first and second moments of the SM time series (Jalilvand et al., 2021)—with the Variable Infiltration Capacity (VIC) LSM using a PBS DA approach. The main objectives of the study are.

1. Remove systematic biases between the LSM and the satellite observations prior to the data assimilation (Section 2.2).
2. Conduct a series of synthetic studies to evaluate the impact of system characteristics such as the frequency of the observations, knowledge of irrigation timing, and presence of noise in the observations on irrigation estimates (Sections 2.3 and 4.1).
3. Quantify the irrigation water use over multiple irrigated and non-irrigated 1 km pixels and evaluate the model performance against in situ observed irrigation data (Sections 2.4 and 4.2).

The article is organized as follows: Section 2 describes the PBS method implementation (Section 2.1), the model calibration prior to the data assimilation (Section 2.2), the suite of synthetic experiments to evaluate the impact

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of different error sources on the DA system (Section 2.3), and the real-world experiments (Section 2.4). Section 3 introduces the study area (Section 3.1) and all the input data to the VIC LSM plus satellite SM data used for assimilation and in situ irrigation observations (Section 3.2). The results of the synthetic and real-world experiments are presented in Sections 4.1 and 4.2, respectively. Section 5 discusses possible shortcomings and improvements in the PBS method by pointing to future satellite missions. Section 6 provides conclusions.

## 2. Methodology

A particle batch smoother (PBS) data assimilation approach is used to quantify irrigation water use (IWU). Following Abolafia-Rosenzweig et al. (2019), we assimilate SMAP-S1 1 km SM observations—that is proven to have irrigation signals (Jalilvand et al., 2021) over the study region—with the VIC (4.2d) LSM (Liang et al., 1994). We conduct a suite of synthetic experiments in an identical twin setup (Abolafia-Rosenzweig et al., 2019; Kumar et al., 2015), where truth SM is created by forcing VIC with precipitation plus a known amount of irrigation, and then evaluate the model performance under different system characteristics such as known or unknown irrigation timing, different frequency of satellite retrievals, and various noise levels. For optimum DA performance, systematic biases between the LSM and satellite retrievals must be addressed (Kumar et al., 2012). We choose an a priori bias correction strategy by calibrating the model against observations (Zhou et al., 2020), as opposed to the common practice of scaling the observations to the model climatology which removes observed irrigation signals (Kumar et al., 2015). Finally, SMAP-S1 SM data are assimilated over multiple irrigated and non-irrigated pixels, and simulated irrigation using the PBS method is evaluated against in situ irrigation observations. The LSM runs at hourly time steps, the spin-up period is Jan. 2013–Dec. 2014, the calibration period is 2015–2019 and the IWU validation is done during 2020 because in situ irrigation data is only available during this year.

### 2.1. Particle Batch Smoother

We use a particle batch smoother (PBS) algorithm to assimilate SMAP-S1 SM observations with the VIC LSM. PBS is a non-sequential extension of a particle filter (PF) in which a series of observations are assimilated within a window of time in a batch to correct model states and fluxes. Dong et al. (2015) showed that using the same observational data, PBS yields significantly better results with a marginal increase in the computational cost compared to PF in temperature assimilation experiments. The main distinctions of PBS compared to the widely used ensemble Kalman Smoother (EnKS) are that: (a) no prior distribution assumptions are made by the PBS (Lu et al., 2016; Yan & Moradkhani, 2016), (b) the model simulation weight can be traced to the corresponding input forcing (here, precipitation plus irrigation) to form the posterior probability distribution function (PDF) of the forcing (Lu et al., 2016), (c) PBS maintains the mass and energy balance of ensemble members, and (d) The PBS is more robust to input uncertainties and prior errors (Dong et al., 2015; Margulis et al., 2015).

Here we describe the PBS implementation. Readers are referred to Abolafia-Rosenzweig et al. (2019), Dong et al. (2015), Margulis et al. (2015) for the detailed presentation of the method.

The precipitation forcing is created by superimposing a random irrigation value during the irrigation season.

$$P^i_{\text{particle}} = P_{\text{obs}} + \text{IRRG}^i(r) \tag{1}$$

where $P^i_{\text{particle}}$ shows the precipitation forcing for each particle, $\text{IRRG}^i(r)$ is a random sample from a uniform distribution of irrigation. Since here we choose to have 99 particles, $P_{\text{particle}}$ would be a $99 \times 1$ vector. The range of irrigation ($r$) is chosen to be between 0 and 12 mm/day during the irrigation season (May-August inclusive) and zero elsewhere. We use two scenarios for adding the irrigation: (a) we assumed that we know the irrigation timing or, (b) the irrigation timing is unknown, and the irrigation is applied continuously. An ensemble of model states evolves in parallel using a forward model.

$$x^i_t = f\left(x^i_{t-1}, u^i_t, b^i\right) + w^i_t \tag{2}$$

where $x^i_t$ is the model state for particle $i$ at time step $t$, $u^i_t$ is the forcing data set, $b^i$ is a vector of time-invariant model parameters, $w^i_t$ is the model error, which is assumed to be normally distributed and, $f$ is the forward model (VIC 4.2 d).

The likelihood of observed, $y$, given particle estimate of, $x$, is calculated based on all observations within the window of length $L$.

$$p\left(y_{t-L+1:t} \mid x^{i}_{t-L+1:t}\right) = \prod_{j=t-L+1}^{t} \frac{1}{(2\pi)^{\frac{n}{2}} \times \det(C_v)^{\left(\frac{1}{2}\right)}} e^{-0.5\left(y_j - x^{i}_{j}\right)^{T} C_V^{-1}\left(y_j - x^{i}_{j}\right)} \tag{3}$$

where $t - L + 1 : t$ represents the window, $n$ represents the number of states or the fluxes that are assimilated (here only SM is assimilated thus, $n = 1$), $C_v$ is the observation error covariance. We used the SMAP-S1 prescribed error for $C_v$. When the window length is $L = 1$ then the observations are assimilated sequentially otherwise they are assimilated in a single batch in each window. The weight for each particle ($w^{i}_{t}$) in each window is calculated by normalizing the likelihood of each particle:

$$w^{i}_{t} = \frac{p\left(y_{t-L+1:t} | x^{i}_{t-L+1:t}\right)}{\sum_{i=1}^{N} p\left(y_{t-L+1:t} | x^{i}_{t-L+1:t}\right)} \tag{4}$$

It is recommended to resample the weights at the beginning of each window to avoid weight degeneration (a situation where a negligible weight is assigned to most of the particles) (Moradkhani et al., 2005; Smyth et al., 2022). We used sequential weights resampling to resample particles based on the weights in the previous window (Gordon et al., 1993; Weerts & el Serafy, 2006). This method gives particles with higher weights a greater chance of propagating their state into the next window as an initial condition (Dong et al., 2015).

Finally, the expected time series of precipitation plus irrigation ($A_{\mathrm{PBS},t}$) will be

$$A_{\mathrm{PBS},t} = \sum_{i=1}^{N} w^{i}_{t} P^{i}_{\mathrm{particle},t} \tag{5}$$

where $N$ is the number of particles (here $N = 99$) and $P^{i}_{\mathrm{particle},t}$ is the particle $i$ forcing at time $t$. The best-simulated irrigation time series is estimated by subtracting observed precipitation ($P_{\mathrm{obs}}$) from the $A_{\mathrm{PBS},t}$.

### 2.2. A Priori Bias Correction

Systematic biases between the model and the satellite observations (from SMAP-S1) must be addressed prior to the analysis step within data assimilation. We acknowledge that both the SMAP-S1 and the model SM values are different indices of soil wetness, and they are not necessarily representative of the real soil moisture, but here we assume that SMAP-S1 is a reasonable observation of surface soil wetness. To remove biases, either observations are scaled to conform to the model's climatology (e.g., through CDF matching), or the model is calibrated against observations. The key consideration is that the bias correction approach should preserve observed irrigation signals while removing biases between simulated and observed SM (excluding biases imposed by unmodeled irrigation). The problem with rescaling methods such as CDF-matching (Reichle & Koster, 2004) is that it might remove unmodeled human-induced signals in SM observations, such as irrigation (Kumar et al., 2015). Therefore, we use the model calibration approach in a batch mode, as suggested by Kumar et al. (2012), where a set of observations are used to estimate the model parameters by minimizing the difference between the model and observations. We choose to only calibrate the model during the rainy season to avoid erasing the unmodeled irrigation process. The objective function for model calibration is minimizing the absolute mean bias between the modeled and the observed SMAP-S1 SM during the rainy season. According to the sensitivity analysis carried out by Zhou et al. (2020), the two main parameters that strongly change the first and second moment of VIC simulated SM are BLKDN (bulk density) and EXPT (the exponent in the Campbell's equation for soil hydraulic properties (Campbell, 1974)). We choose to fit the BLKDN (variation range from 1,200 to 1,400 [kg/m$^3$]) and EXPT (variation range from 3 to 20 [−]) for SM model calibration. The ranges are chosen based on the soil physical properties of the case study obtained from SoilGrid 250 m (www.openlandmap.org) data (Hengl et al., 2017).

### 2.3. Synthetic Experiment Sensitivity Analysis

We use a suite of synthetic experiments to evaluate the sensitivity of the DA system to potential error sources. The DA system is evaluated by introducing sources of error sequentially and combined as shown in Table 1. The synthetic truth was created by forcing the VIC model with a known amount of irrigation plus observationally based precipitation. Synthetic observations were generated by sampling the truth simulation at different time steps namely, daily, SMAP-S1 frequency, and every 12 days (SMAP-Sentinel 1 A only).



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**Table 1**
*Summary of the Synthetic Experiment Results*

| Exp No | Noise | Observation frequency | Irrigation timing | Accumulated irrigation bias (%) | Irrigation RMSE (mm/hr) | Irrigation $R^2$ |
|---|---|---|---|---|---|---|
| 1 | No noise | Daily | known | +0.51 | 0.014 | 0.993 |
| 2 | | | continuous | +16.67 | 0.141 | 0.394 |
| 3 | | SMAP-S1 frequency (6 days) | known | +0.41 | 0.026 | 0.977 |
| 4 | | | continuous | +22.93 | 0.143 | 0.394 |
| 5 | | 12 days (Sentinel 1A-only) | Known | +0.712 | 0.029 | 0.9735 |
| 6 | | | continuous | +26.42 | 0.154 | 0.318 |
| 7 | With Noise20 realizations mean | Daily | Known | +0.09 | 0.017 | 0.990 |
| 8 | | | continuous | +18.25 | 0.143 | 0.381 |
| 9 | | SMAP-S1 frequency (6 days) | Known | −3.05 | 0.043 | 0.943 |
| 10 | | | continuous | +24.93 | 0.150 | 0.341 |
| 11 | | 12 days (Sentinel 1A-only) | Known | −7.43 | 0.043 | 0.953 |
| 12 | | | continuous | +25.71 | 0.161 | 0.254 |

To evaluate the impact of noise in the satellite retrievals we run experiments by imposing random 0-mean Gaussian noise to the synthetic observations with a standard error of 0.05 $m^3/m^3$ (unbiased root mean square error (ubRMSE) of SMAP-S1; Das et al., 2019). We run each of these observational-noise experiments (Exp No 7–12 in Table 1) 20 times (20 data assimilation experiments with unique sets of random noise imposed on synthetic observations) and report the mean irrigation of 20 realizations as the final simulated irrigation of that experiment. We further evaluate DA performance when irrigation timing is known or unknown. In the case of unknown irrigation, we applied irrigation continuously (all day, every day) throughout the irrigation season (May-August inclusive). This is a valid assumption in the regions with small agricultural land parcels (e.g., a few acres) because a fraction of the SMAP-S1 1 km pixel could be always under irrigation (see Figure 2). The range of plausible irrigation applications varies based on the climate, crop type, and human choices. We evaluate the sensitivity of DA performance to changing the range of superimposed irrigation during the irrigation season.

### 2.4. Real World Experiment

We perform real-world data assimilation experiments over multiple irrigated (IR) and non-irrigated (nIR) pixels in our study region. According to our previous study (Jalilvand et al., 2021) high variability (i.e., high mean absolute deviation (MAD)) in the SMAP-S1 SM time series during the irrigation season is an indicator of irrigation signal being captured by the satellite observations. Therefore, four irrigated pixels with a high MAD value and the cropland land use type that are covering an irrigation channel (for which in situ irrigation data is available) are selected as the study IR pixels. Simulated irrigation is divided by the fraction of irrigated area in each pixel obtained from the land use map of the region (Mousivand et al., 2020). We run the model over nIR pixels to estimate the amount of false irrigation over the non-irrigated area. Thereby, seven nIR pixels located nearby the irrigated area (within 36 km SMAP radiometer footprint that also covers irrigated pixels) all with low MAD value and land use/landcover type of range land or rainfed agricultural are chosen as the nIR pixels (nIR 01). To test the impact of possible irrigation signal leakage from the IR pixels to the neighboring nIR pixels we choose to run the model over four other nIR pixels that are located further away from the irrigated pixels (outside of the 36 km base SMAP radiometer footprint that covers the irrigated area) (nIR 02). Pixel locations, underlying MAD value, and locations of irrigated channels are shown in Figure 1.

The LSM simulation forced only with meteorological conditions (i.e., no irrigation) is termed the open-loop (OL) run, the simulation calibrated against SMAP-S1 SM observations and forced with meteorological conditions is used as the calibrated simulation (CAL). The weighted average of simulations forced with precipitation plus a random amount of irrigation used as the particles is termed the PBS. PBS simulations are compared with the observed irrigation (Section 3.2.3). Simulated irrigation over the irrigated and non-irrigated pixels are tested for the difference of medians using the Wilcoxon right-tale rank-sum test.

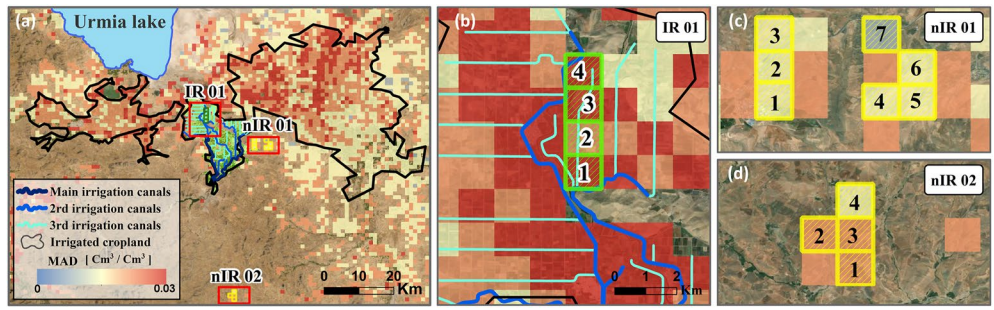



**Figure 1.** | Overview of the study area. Irrigated (IR) and non-irrigated (nIR) pixels are chosen based on the Soil moisture mean absolute deviation (MAD) value (higher MAD for the IR and lower MAD for the nIR), the location of the irrigated cropland area, and landcover/land use type. (a) The entire case study and location of selected IR and the nIR pixels (boxes outlined in red), (b) irrigated pixels (IR) covering one of the irrigated channels for which in situ data is available, non-irrigated pixels within the 36 km of irrigated pixels (nIR 01) and, (d) non-irrigated pixels (nIR 02) farther down the irrigated area (outside of the 36 km base SMAP radiometer footprint that covers the irrigated area).

## 3. Data and the Study Area

### 3.1. Study Area

Urmia lake has experienced a rapid decline in water volume in the past two decades due to anthropogenic activities and climate change. On the human part, most of the drying is blamed on the expansion of irrigated agriculture around all the tributaries to the lake (AghaKouchak et al., 2015; Parsinejad et al., 2022). Indeed, the agricultural water demand is more than all the surface water input to the lake combined (Schulz et al., 2020). To stop the drying of the lake different measures have been taken by the Urmia Lake Restoration Program

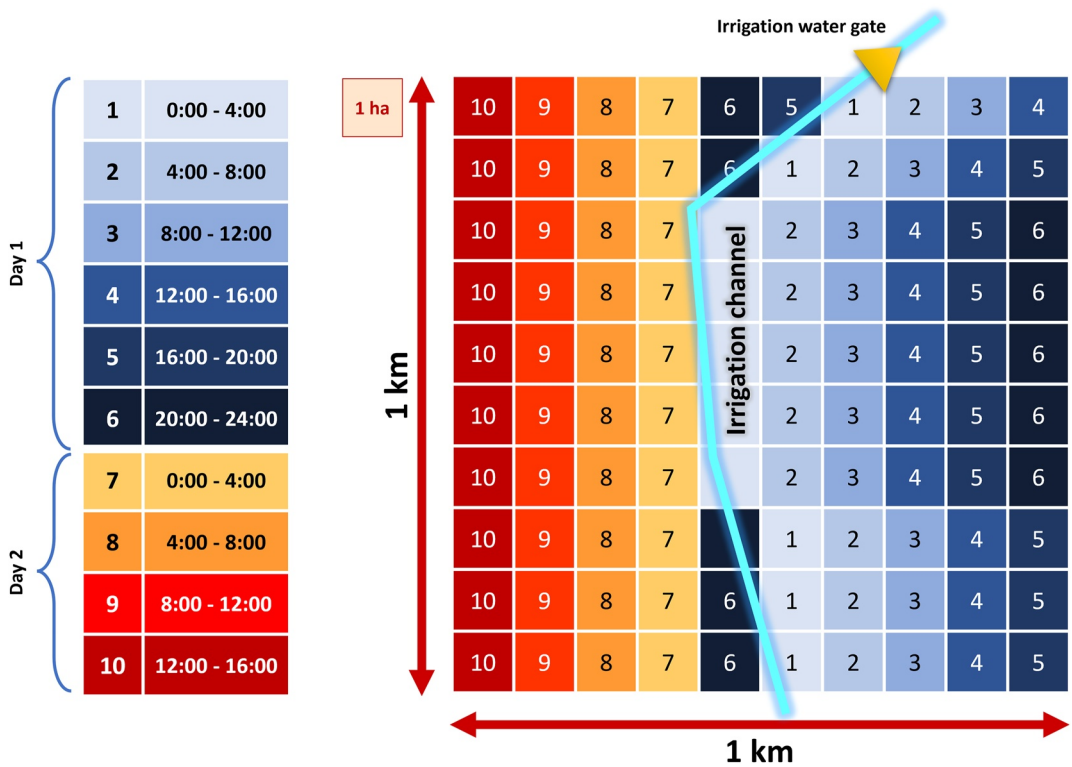



**Figure 2.** Schematic irrigation scheduling in the Mahabad region within a 1 km pixel. 1 ha land parcels on the left of the irrigation channel that are closer to the channel (blocks denoted by number 1) are the first to receive water allotment. Then after 4 hr, the next group of plots (denoted by number 2) receives their water allotments, and this will continue until all the fields on both sides of the channel are irrigated.



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(ULRP) including a 40% reduction in irrigation water use that specifically targeted the agricultural sector (Sima et al., 2021). Later studies show that a further 10% reduction is needed to buffer the impact of climate change (Schulz et al., 2020). 60% of lake Urmia's surface water comes from four southern rivers (Zarineh Rood, Simineh Rood, Mahabad, and Gadarchay) that pass through three major irrigated agricultural regions (Miandoab, Mahabad, and Naghadeh). Mahabad has a fully established irrigation network downstream of the Mahabad Dam and gravity irrigation is the common irrigation practice in the region (Shadkam et al., 2016). The main crops cultivated in Mahabad are wheat, barley, and alfalfa, which are usually irrigated between May to the end of August (Zaman et al., 2016).

## 3.2. Data

### 3.2.1. VIC Model

The Variable Infiltration Capacity (VIC) is a distributed LSM (Liang et al., 1994) that simultaneously solves surface water and energy budgets at each grid cell independently. We choose to work with VIC 4.2d version which has the ability to simulate partial vegetation cover and photosynthesis and can input time-varying vegetation parameters. We run the model at EASE-GRID 2.0, 1 $km^2$ grid (Brodzik et al., 2012), and hourly time steps. Three soil layers are considered in the model implementation with 0.1 m, 0.9 m, and 1.976 m depth from the top to the bottom. Despite the depth difference between the first layer and the SMAP nominal sensing depth (5 cm), we chose to not change the default model structure fundamentally as the influence of the layer depth difference is likely small compared to the model component parameterizations that dominate systematic differences with observation (Shellito et al., 2020). Moreover, by keeping the default soil layer setup, the optimal soil parameter remains relevant to the community that uses the model (Pinnington et al., 2021). This is a common practice in many other studies that assimilated satellite soil moisture data with VIC (Lievens et al., 2016; Xia et al., 2012a, 2012b; Zhou et al., 2020) or other LSMs (Pinnington et al., 2021; Rahman et al., 2022; Rodríguez-Fernández et al., 2019; Santanello et al., 2016). On the other hand, The SMAP sensing depth can vary based on the soil moisture content from ~5 cm (when the soil is saturated) to more than 15 cm (in the drier soil) according to a recent study by Feldman et al. (2022).

#### *3.2.1.1. Atmospheric Forcing*

The atmospheric forcing inputs are hourly precipitation, minimum and maximum temperature, and wind speed. Daily Integrated Multi-satellitE Retrievals for GPM (IMERG) final run V06 product (Huffman et al., 2019) at 0.1° resolution is used as the precipitation forcing. Wind speed and maximum and minimum air temperature data are obtained from the GLDAS Noah Land Surface Model L4 3-hourly 0.25° V2.1 (GLDAS_NOAH025_3H) (Rodell et al., 2004). Following Abolafia-Rosenzweig et al. (2019), the wind and the precipitation datasets are bicubically resampled while the temperature data set is disaggregated bilinearly to the EASE-GRID 2.0, 1 $km^2$ grid. Hourly disaggregation of daily precipitation is performed by equally dividing the daily precipitation throughout the 24 hr and the hourly wind and air temperature are created by replicating the same value for each 3-hr interval. We acknowledge that downscaling wind and temperature data is an error prone process and can play a role in simulated errors at the 1 km resolution, but to the authors' knowledge, the GLDAS data was the best option available in this region. Furthermore, our case study is located in an area of low topographic complexity which tends to have lesser spatial heterogeneity in temperature and wind. We use the finer resolution product (IMERG) for precipitation—which is the primary driver of SM variations—to reduce these errors.

#### *3.2.1.2. Vegetation Forcing*

Vegetation forcing variables—LAI, Albedo, and vegetation fraction—are from MODIS products. 8-Day, 500 m MODIS combined Leaf Area Index (MCD15A2H, V006) is used for the LAI forcing (Myneni et al., 2015), daily 500 m MODIS combined Albedo (MCD43A3, V006) is used for the Albedo forcing (Schaaf & Wang, 2015), and 16-Day, 500 m EVI (MYD13A1, V006) is used to calculate the vegetation fraction (Didan, 2015). All the data are linearly interpolated in time and space to make a daily, 1 $km^2$ (EASE-GRID 2.0) vegetation forcing.

#### *3.2.1.3. Soil Parameters*

VIC soil parameters are generated from the 250 m SoilGrid data (Hengl et al., 2017) using soil empirical equations (Saxton & Rawls, 2006) at 5 $km^2$ spatial resolution. This data is disaggregated to the model grid using the nearest neighbor approach.



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#### 3.2.2. SMAP-Sentinel 1

We use an updated version of SMAP-S1 1 km SM (Das et al., 2019, 2020) with dynamic vegetation water content that has irrigation signals in both the first and second moment of the SM time series (Jalilvand et al., 2021). The Sentinel 1A AM pass is merged with the same morning or the previous afternoon SMAP crossing to ensure the stability of SM content between two successive radiometer and radar retrievals. The data is further filtered at the pixel level based on time differences between the Sentinel 1A and SMAP retrievals.

#### 3.2.3. In Situ Irrigation Data

Since 2020, ULRP has installed multiple flowmeters on third-order channels in the Mahabad irrigation network. Each of the flowmeters measures the water allotment of roughly 3 $km^2$ of the irrigation cropland. The volume of the water, the area that is irrigated, and the start and the end of each irrigation event are recorded. Each 1 ha of land receives 4 hr of irrigation, in order of distance to the irrigation channel as shown in Figure 2. The depth of irrigation (mm/hr) at each irrigation event is obtained by dividing the volumetric irrigation by the area irrigated and number of irrigation hours.

## 4. Result

### 4.1. Synthetic Experiment

The results from the synthetic experiments (summarized in Table 1) show that prior knowledge of irrigation timing has a significant impact on the accuracy of estimated irrigation, and assuming continuous irrigation tends to cause overestimates. Indeed, in most of the simulations with known irrigation timing, we obtain a bias of less than 1%, when the irrigation schedule is unknown, the average bias between simulated and observed irrigation increases to 22% and RMSE increases by 5–10 times. Moreover, when irrigation timing is unknown, the model fails to capture the temporal pattern of irrigation ($R^2$ at hourly timeseries drops to less than 0.40). Imposing noise on synthetic observations resulted in less accurate irrigation; although, more frequent observations can help reduce the noise impact as they provide more irrigation signal rather than noise (Abolafia-Rosenzweig et al., 2019). The uncertainty in synthetic observations is shown by the red band in the irrigation plots in Figure 3 which are anticorrelated to the frequency of the observations in each window. This means the possible range of estimated irrigation can significantly increase (model uncertainty increases) when the temporal resolution of the observations, for example, satellite SM, is low. This error can also propagate to the next window as potentially inaccurate particles with high weights have a high probability of propagating to the next window in the resampling process.

In the zero-noise experiments, the bias is persistently positive (irrigation is overestimated) in both known and unknown irrigation timing scenarios (Exp. 1–4). When random noise is imposed and irrigation timing is known (Exp No 7, 9, and 11) the bias is nearly zero or negative. These biases are partially attributable to missing some irrigation events due to lower observation frequency.

### 4.2. Real-World Experiment

The model underestimates seasonal cumulative irrigation at the irrigated pixels by −18.6% (average of four IR pixels) when irrigation timing is known and overestimates irrigation by 5.5% when the irrigation timing is unknown (Figure 4). The underestimation in the known irrigation timing simulation is higher for the real-world experiment than in the corresponding synthetic experiment (Table 1, Exp. 9) which is likely partially attributable to losing part of the irrigation signal in the SMAP-S1A SM retrievals which have imperfect accuracy. The coarse spatial resolution of SMAP-S1 1 km observations relative to plot scale agriculture, simplified LSM physics, and uncertainty in LSM input data are among other factors contributing to the prediction errors. Furthermore, in situ measurement of irrigation from surface water may have uncertainty because a portion of water is lost in the delivery system after the measuring station (Shadkam et al., 2016). Moreover, there are reports of illegal (and un-recorded) surface water extractions from the irrigation channels in the region. Thus, in practice, the amount of water that is used for irrigation in downstream fields is lower than observations, potentially contributing to the underestimation by PBS.

Simulations using unknown irrigation timing result in more accurate accumulated irrigation relative to the corresponding synthetic experiment, which is counter intuitive, but this is attributable to compensatory errors (i.e., the

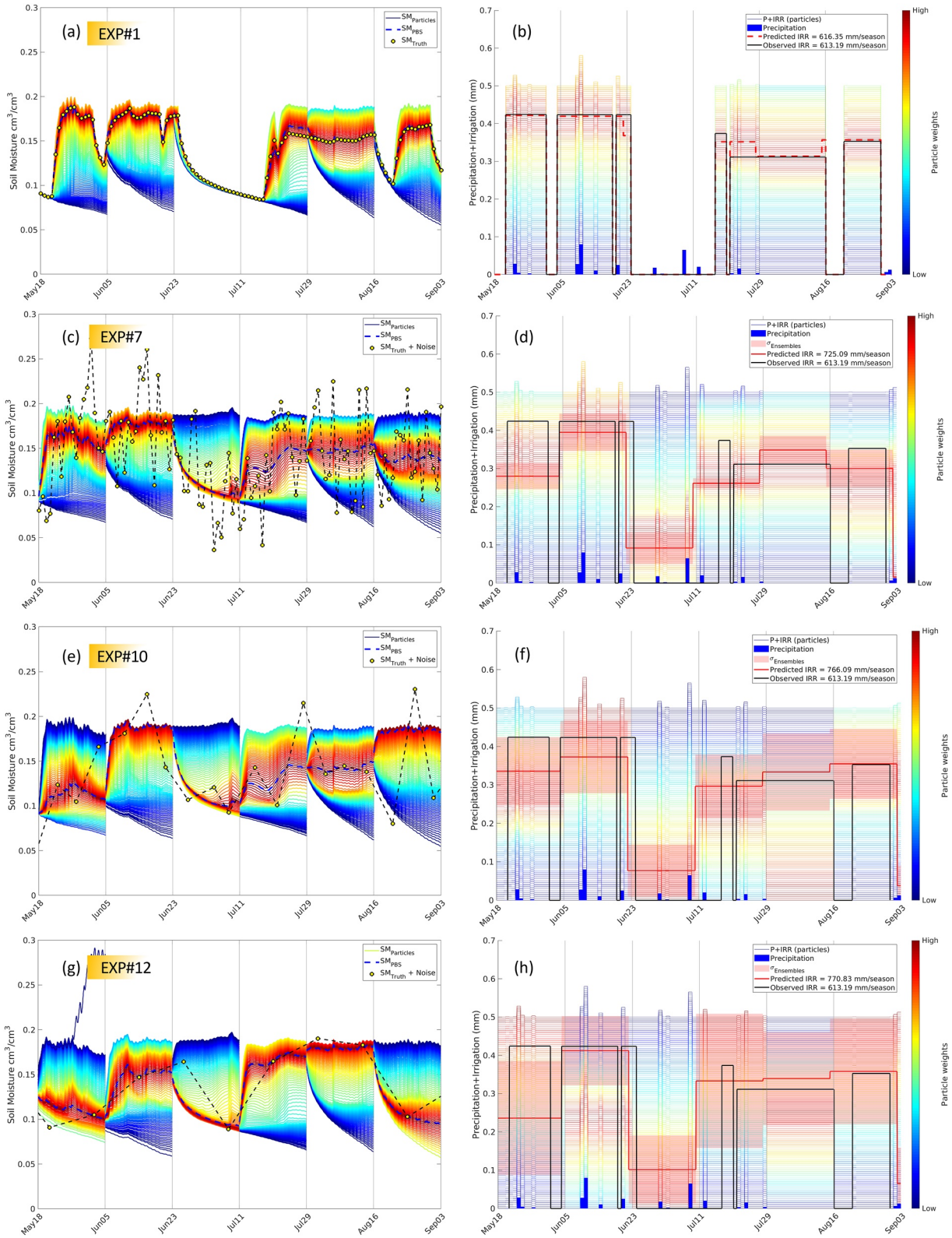



**Figure 3.**



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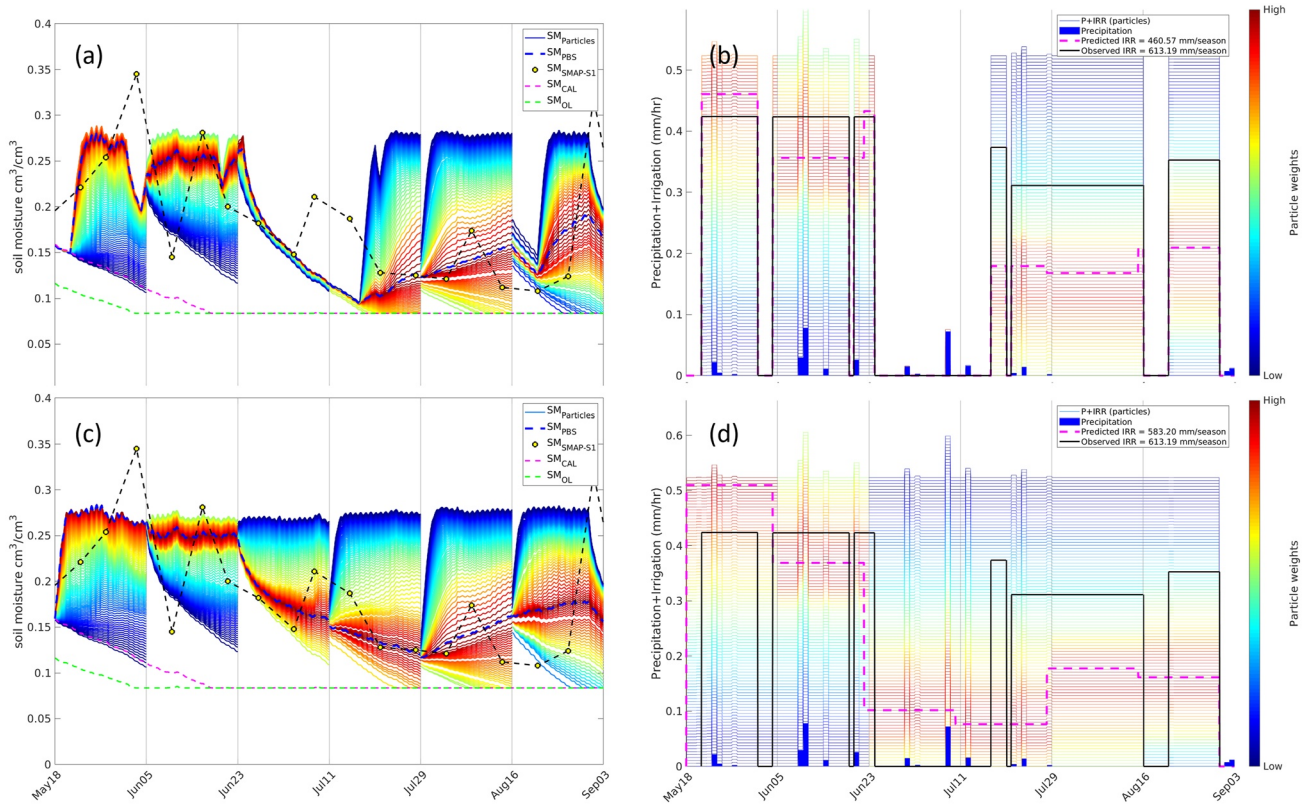



**Figure 4.** | Assimilating SMAP-Sentinel 1 (SMAP-S1) soil moisture (SM) data using a particle batch smoother approach over irrigated pixels. (a) SM timeseries of 99 particles colored by the weights assigned based on the proximity to SMAP-S1A SM observations (dashed-black line and yellow dots) and the best-estimate of SM (dashed blue line) when irrigation timing is known, and (c) when it is unknown. The dashed magenta and green line in the SM time series shows the calibrated and open-loop runs. (b) The corresponding time series of IRR particles constructed by tracing the weights to the particle forcing and the best estimate of irrigation (dashed magenta line) when irrigation timing is known and (d) when it is unknown.

right answer for the wrong reason). In the continuous simulation, we still mostly underestimate irrigation during the irrigation season (especially from June to the end of the irrigation season), but the estimation of some irrigation amount during the gap in the irrigation application compensates for underestimation in the irrigation period.

The magenta and the green dashed line in the SM plots of Figures 4a and 4c show the OL and CAL runs that demonstrate how model calibration removes the systematic bias at the beginning of the simulation and how PBS DA introduces the unmodeled irrigation process and improves the simulated SM relative to SMAP-S1 observations.

We estimate near-zero irrigation in non-irrigated pixels (Figure 5) far away from the irrigated pixels (outside of the 36 km radiometer base SMAP product footprint covering the irrigated area). Estimated irrigation in the nIR pixels that are located within the same base radiometer SMAP product as the irrigated pixels is significantly higher than the nIR pixels far away from the IR pixels, which is likely due to the leakage of the irrigation signal from the IR pixels to the nearby nIR pixels and indicate room for improvement in the downscaling approach. Yet the amount of irrigation at these neighboring nIR pixels (nIR01) is significantly lower than the IR pixels (Figure 6).

**Figure 3.** | Synthetic experiment shows that knowing irrigation timing and having more frequent SM retrievals significantly improves simulated irrigation. Left-hand plots show particles ($\boldsymbol{SM}_{particles}$) colored based on proximity to the synthetic truth SM ($\boldsymbol{SM}_{truth}$) shown as a dashed black line and yellow circles, and the dashed blue line shows the weighted average of all particles ($\boldsymbol{SM}_{PBS}$). Right-hand plots show the corresponding irrigation time series where the weighted average irrigation simulation is shown as the solid red line (the average of 20 realizations with unique random noise is considered the best estimate irrigation) and observed irrigation is the solid black line, the light red band represents one standard deviation of 20 simulations with random noise. Synthetic experiments are conducted with different assumptions (a and b) Exp No 1: ideal situation; synthetic truth at daily timesteps, without adding noise and knowing the irrigation timing, (c and d) Exp No 7: synthetic truth at daily timestep with random noise added and not knowing the irrigation timing, (e and f) Exp No 10: synthetic truth resampled at the SMAP-S1 frequency, with random noise added and not knowing the irrigation timing and, (g and h) Exp No 12: synthetic truth resampled at the SMAP-Sentinel 1 A only frequency (every 12 days), with random noise added and not knowing the irrigation timing.



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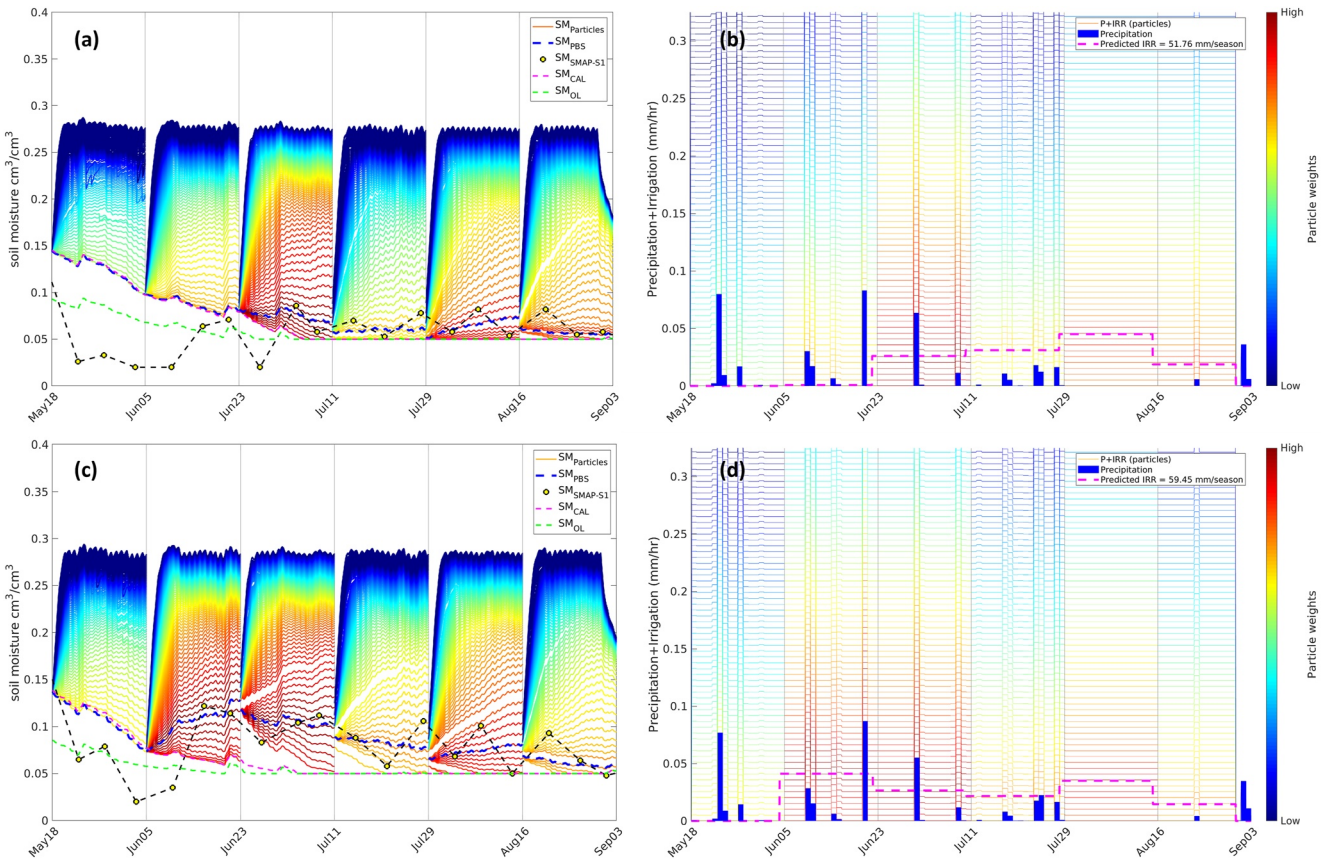



**Figure 5.** | Near-zero irrigation is estimated over non-irrigated pixels far away from the irrigated area. (a) SM time series of 99 particles colored by the weights derived from the proximity to the SMAP-Sentinel 1 soil moisture data (black dashed-(yellow) dotted line) and the best estimate of SM (dashed blue line) over the non-irrigated pixel 4 (nIR02) and (c) pixel 1 (nIR02) (see Figure 1 for the location of the pixels). The dashed-magenta and green line in (a and c) show the calibrated simulation and open-loop run, respectively. (b) The corresponding time series of IRR constructed by tracing the weights to the particle forcing and the best estimate of irrigation (dashed magenta line) in the non-irrigated pixel 4 (nIR02), and (d) pixel 1 (nIR02).

Calibration theoretically removes systematic biases between LSM and observed SM, excluding differences caused by unmodeled processes such as irrigation. Figure 7 shows that calibration alone increases agreement between simulated and SMAP-S1 observed SM over the irrigated pixels (lower RMSE by nearly 10% on average) compared to the OL run during the irrigation season. The PBS DA accounts for the remaining 50% reduction in

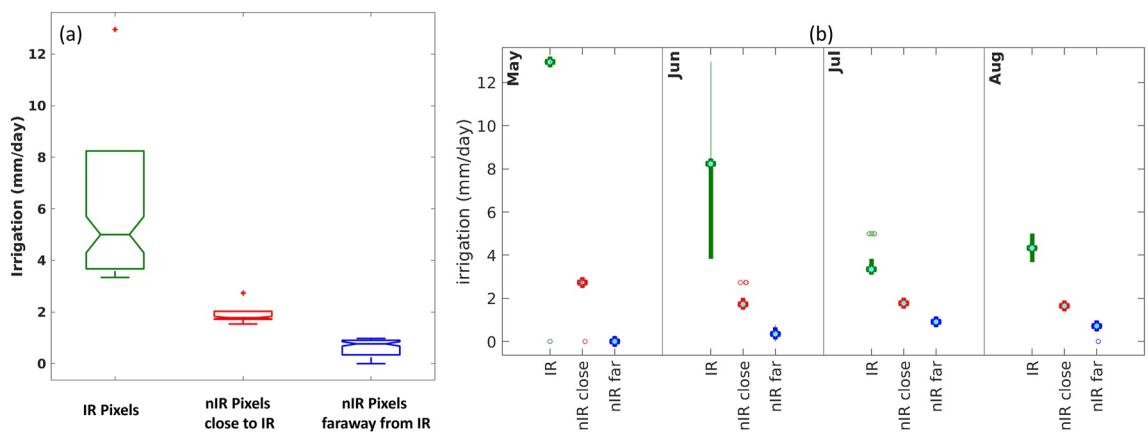



**Figure 6.** | Significantly higher irrigation is obtained over the irrigated pixels (green) compared to nIR pixels close to the irrigated area (red) and non-irrigated pixels far away from the irrigated area (blue), (a) during the irrigation season and, (b) during each month of the irrigation season.



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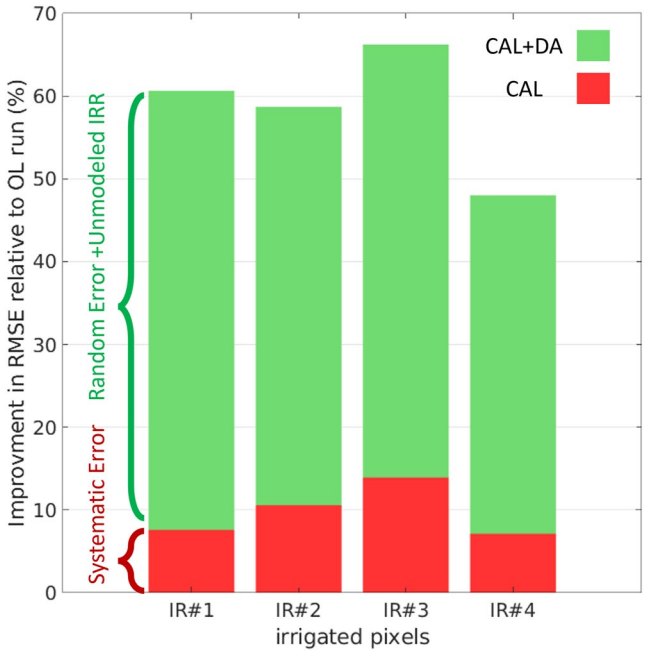



**Figure 7.** Reduction in soil moisture RMSE comparing simulated and SMAP-Sentinel 1 observations relative to the open-loop run after model calibration (red bar) and after both calibration and data assimilation (red and green bar) during the irrigation season.

RMSE by introducing the unmodeled irrigation forcing and contributing to the reduction of random noise.

## 5. Discussion

### 5.1. The Impact of Less Frequent Retrieval on the Estimated Irrigation

The impact of less frequent SM retrievals on the SM estimation is twofold: (a) it is more likely that irrigation events are missed especially over the arid and semi-arid areas where the SM memory is low (McColl, Alemohammad, et al., 2017; McColl, Wang, et al., 2017). This favors significant underestimation of irrigation; and (b) having multiple retrievals in one smoothing window can average out an outlier observation, however having an outlier (e.g., a very noisy observation) in a sparse SM time series at either end of the SM spectrum (saturated or residual SM) can lead to choosing the wettest or the driest particle in that window and consecutively over-or underestimation of the irrigation, respectively.

### 5.2. Possible Problems With an Unknown Superimposed Irrigation Range

The maximum superimposed irrigation value can be among the unknowns in many regions. When SM observations are close to saturation, the PBS model will favor the wettest particles. Therefore, if 10 mm irrigation results in the soil saturation in one window, choosing a 40 mm/day as the maximum of the supplement irrigation can result in a 4 times higher irrigation estimation in that window as SM cannot increase beyond saturation. A similar situation can be seen in the first window of Figure 4, where a saturated observation resulted in the selection of the wettest particle and the overestimation of irrigation in that window. Hence, this methodology is not appropriate for irrigation quantification over regions that maintain saturated soils via irrigation.

### 5.3. Posterior Bias Correction

The false irrigation estimate at non-irrigated pixels can be treated as the model's positive bias and be removed from the irrigation estimation over the irrigated pixels. However, irrigation should be the only difference between the IR and the nIR pixels, otherwise the posterior bias correction might remove the true irrigation signal. Jalilvand et al. (2019) argued that choosing an adjacent nIR pixel for the posterior bias correction can maximize the climate similarity as the estimated irrigation at the nIR pixel may exclusively cause by the model bias rather than a different precipitation pattern. Brombacher et al. (2022) used a hydrological similarity concept (van Eekelen et al., 2015) to compare actual evapotranspiration ($ET_{act}$) of an IR pixel with an average $ET_{act}$ of natural (nIR) pixels to calculate the $ET_{act}$ caused by the irrigation (incremental ET). They define hydrological similarity based on a set of features derived from soil texture, DEM, reference ET, and precipitation datasets. The same concept can be adopted in future studies for the posterior bias correction of irrigation estimated by the PBS method.

### 5.4. Moving Forward and Future Consideration

#### 5.4.1. Assimilating Other High-Resolution Satellite Soil Moisture Products

High-resolution (≤1km) satellite soil moisture products such as daily Sentinel1, 1 km SM data for Europe (Bauer-Marschallinger et al., 2018), BEC SMOS L4, 1 km SM data for Europe (Barcelona Expert Center, 2020) or the planet labs 1 km experimental SM data set for the Mediterranean terrestrial region (Schellekens et al., 2022) can potentially be assimilated using the PBS approach to quantify the IWU. The other good candidates are machine learning-based and hybrid satellite-model products such as downscaled 1 km SMAP product using the ensemble learning approach (Abbaszadeh et al., 2019, 2021) or the 30 m satellite and hyper-resolution land surface model data (Vergopolan et al., 2021, 2022) over the US, provided they contain the irrigation signal. The



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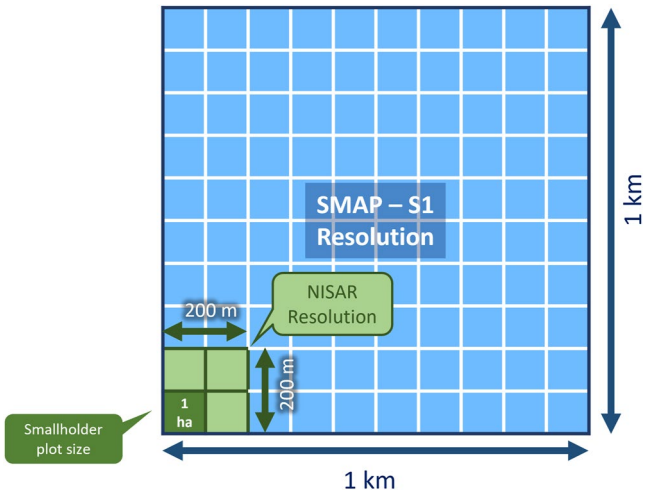



**Figure 8.** | The future NASA-ISRO Synthetic Aperture Radar (NISAR) high-resolution soil moisture data can significantly improve the irrigation simulation at the plot scale using the particle batch smoother approach.

spatial mismatch between satellite and smallholder agricultural plots is likely to remain an issue in most of these products, though.

Future satellite missions such as the NASA-ISRO Synthetic Aperture Radar (SAR) mission (NISAR) that is planned to lunch in 2024, with 200 m spatial resolution (Figure 8) and retrieval frequency of 6 days can significantly improve the irrigation simulation and quantification as it can observe SM changes at the scale relevant to the most smallholder plots globally. Indeed, the non-irrigated area within the 1 km$^2$ SMAP-S1 pixel can contribute to SM retrieval and alleviate the irrigation signal. This fact is confirmed by estimating significantly higher irrigation in the synthetic experiment (where it is assumed that the satellite SM product backs out the entire irrigation signal), compared to the corresponding real-world experiment.

### 5.4.2. Smart Particles

Particles can be designed to not superimpose irrigation unless an irrigation event in an irrigated pixel is detected by the satellite. In our previous study (Jalilvand et al., 2021), we demonstrate that the IR pixels show a significantly higher variation (MAD value) in the SM during the irrigation season compared to the nIR pixels. Thus, the satellite SM MAD value can be used to filter non-irrigated pixels prior to the DA. Furthermore, irrigation can be added to the precipitation forcing of the IR pixels when a positive increment in the SM, which is not caused by precipitation, is observed. One approach to identify this is to compare the soil moisture/radar backscatter time series at the plot scale to the regional scale, as irrigation changes soil moisture at a plot scale while atmospheric forcing drives changes at a larger scale (Bazzi et al., 2022; Zappa et al., 2022). Alternatively, comparing SM simulation from a model without an irrigation module to a high-resolution soil moisture product that contains the irrigation signal can help detecting an irrigation episode (Le Page et al., 2023). However, infrequent satellite observations, dense vegetation, and low irrigation rate (e.g., drip irrigation) are among the factors that might limit the applicability of these methods.

### 5.4.3. Multi-Variate Data Assimilation

Although univariate assimilation of soil moisture can provide irrigation signal to LSM simulations, this technique is inherently limited by the sensing depth of remotely sensed soil moisture retrievals (topsoil layer). Future analyses can consider assimilating both soil moisture and other remotely sensed hydrologic variables that have irrigation signals (Abolafia-Rosenzweig et al., 2019; Deb et al., 2022). In particular, multi-variate assimilation of SM and ET has already been proven to provide benefits for soil moisture and agricultural drought simulations (Gavahi et al., 2020; Xu et al., 2020). Similar improvement is expected in the irrigation simulation as simultaneous ET and SM assimilation provides unique information on the plant's soil water use throughout the entire rooting depth and not just the topsoil.

## 6. Conclusion

Irrigation is the largest human-engineered modification in the global water budget that is largely unknown and poorly represented in land surface and hydrological models. Irrigation can directly change the SM which can be detected through microwave (MW) remote sensing. In this study, we used a particle batch smoother (PBS) approach to assimilate the SMAP-S1 1 km SM data with the VIC (4.2d) land surface model to quantify the irrigation water use. The main findings are.

1. By calibrating the model against SMAP-S1 observations during the rainy season we reduced systematic biases between the LSM, and satellite observed SM. Unlike commonly used rescaling approaches such as CDF-matching, this bias correction strategy allows for preserving and incorporating observed irrigation signals through data assimilation.
2. We evaluate the impact of known error sources by conducting a suite of synthetic experiments. Results showed that the knowledge of the irrigation timing can significantly improve the irrigation simulation. Moreover, the frequency of observations can help reduce the impact of random noise in the observations.



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3. The method is applied to multiple irrigated pixels for which in situ irrigation data was available. We underestimate the irrigation by an average of 18.6% which can be due to losing part of the irrigation signal in the SMAP-S1 1 km SM retrievals. The presence of non-irrigated areas in 1 km SMAP-S1 SM data can also alleviate the irrigation signal from smallholder plots.
4. We estimate significantly lower irrigation over the non-irrigated pixels compared to the irrigated pixels. The false irrigation estimated at the nIR pixels within the same SMAP 36 km base pixel as the IR pixels, was significantly higher than the far away nIR pixels outside of the 36 km SMAP base footprint. This supports the hypothesis that irrigation signals leak from irrigated pixels to nearby non-irrigated pixels.
5. If the farmers provide their irrigation schedule (i.e., start and end of each irrigation event), we can theoretically quantify the amount of seasonal irrigation with 3% accuracy and $R^2 = 0.95$ (Table 1, Exp No. 9) using a weekly remotely sensed SM product that maintains the majority of irrigation signals (potentially NISAR 200m SM data).
6. Improved high-resolution and accurate observations possibly through future satellite missions (e.g., NISAR) would reduce the impact of leakage errors and can help to circumvent the issue of unknown irrigation schedules.

## Data Availability Statement

NASA SMAP/Sentinel-1 L2 Radiometer/Radar 30-Second Scene 3 km EASE-Grid, V03 is used for soil moisture data (Das et al., 2020). Daily Integrated Multi-satellitE Retrievals for GPM (IMERG) final run V06 product (Huffman et al., 2019) is used for the precipitation data. Wind speed and maximum and minimum air temperature data are obtained from the GLDAS Noah Land Surface Model L4 3-hourly 0.25° V2.1 (GLDAS_NOAH025_3H) (Rodell et al., 2004). 8-Day, 500 m MODIS combined Leaf Area Index (MCD15A2H, V006) is used for the LAI forcing (Myneni et al., 2015), daily 500 m MODIS combined Albedo (MCD43A3, V006) is used for the Albedo forcing (Schaaf & Wang, 2015), and 16-Day, 500 m EVI (MYD13A1, V006) is used to calculate the vegetation fraction (Didan, 2015). Soil parameters are generated from the 250 m SoilGrid data (Hengl et al., 2017). Figures and maps were made with Matlab 2021b under academic/coursework license for members of Michigan State University.

**Acknowledgments**
The authors thank the Urmia Lake Restoration Program for providing financial support for a part of this research and the in situ irrigation data of Mahabad plain. Dr. Behdad Chehrehnagar and Ali Hajmoradi for facilitating the in situ data collection process. NASA SMAP Mission Science team (Grant 80NSSC21K0907) and NISAR Mission Science Team (Grant 80NSSC21K1181) for providing financial support for a part of this research.

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